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Similarity and Classification Analysis in Turkish Question Generation of AI Tools According to Bloom's Taxonomy

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Abstract

One of the most striking features of artificial intelligence (AI) applications is the potential for chatbots to generate human language, with their capabilities increasingly harnessed across a wide array of domains in place of human-produced content. This paper aims to project the holistic view of the intersection of AI chatbots ChatGPT, Gemini, and Copilot with pedagogical frameworks, especially their potential for generating educational questions within the context of Turkish. In this context, it investigates how effective such chatbots are in question generation concerning the Revised Bloom's Taxonomy by comparing AI-generated questions against those crafted by human experts. The approach involves using the expert's questions as reference points to analyze similarities and assess the ability of the AI questions to classify both their output and the reference questions.

Keywords: *Question Classification, ChatGPT, Gemini, Microsoft Copilot, Bloom's Taxonomy*

Introduction

The oldest records of human language transitioning from symbolic commands to complex patterns date back 50,000 years (Klein, 2017). In contrast, the transition of machine language from the symbolism of 1s and 0s to complex patterns dates to the 1950s (Jones, 1994). The vast gap between

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the long development of human language and the recent progress of complexity by machines imposes upon us the need to be concerned with the intricacies of language acquisition. Beginning from 2000, technologies in deep learning and big data accelerated Artificial Intelligence development. These developments have made machines learn from data via language representation and perform human-like interactions through Natural Language Processing techniques (Tohma and Kutlu, 2020). However, for a machine to really understand the language and its intricacies, multiple approaches are required. Chomsky, a linguist and thinker, compared human and machine-produced texts, stating that "the correct explanations of language are complicated and cannot be learned just by marinating in big data" (Chomsky et al., 2023).

Education as a scientific framework and Bloom's Taxonomy as a benchmark can be used to understand the distinction between AI-generated and human-produced language. Bloom proposed the 'taxonomy of educational objectives,' which has been key in shaping question formulation and attainment of competencies in education (Bloom, 1956). Criticism prompted its revision and update by Anderson and Krathwohl in 2001 to fit in with changing pedagogy (Anderson and Krathwohl 2001). Revised Bloom's Taxonomy (RBT) examines cognitive processes and organizes them into six categories: remember, understand, apply, analyze, evaluate, and create. However, the taxonomy's complexity and diversity can pose significant challenges in expressing, classifying, implementing, and evaluating objectives within its framework (Bümen, 2006). Verbs play a significant role in developing common language related to competencies and in the formulation of appropriate questions among the categories of the taxonomy. A study conducted by Stanny into the English verbs in this respect analyzed lists of the verbs' appearance frequency published in the top 30 sites in Google searches, ranked with relation to their appearance in RBT (Stanny, 2016). The same study also stated that verbs sequenced by RBT levels were still inadequate to account for learning differentiation and academic levels based on among other factors the flexibility of the language, the multiple connotations of the same verb used in different contexts.

In the evaluation of educational and scientific consistency in artificial intelligence generation, some studies use the Revised Bloom's Taxonomy as a benchmark. A recent study assessed the scientific acceptability level for multiple-choice questions (MCQs) developed automatically by ChatGPT. Comparative results from this study demonstrate similar quality levels between MCQs generated by ChatGPT and those by subject matter experts (Kıyak, 2023). Another study that sought to determine the possible impact of controllable text generation using Large Language Models (LLMs) on education stated that it could be an effective tool. However, the experiment showed instances where LLM repeated similar queries despite variations in the applied control parameters (Elkins et al., 2023). In another study focusing on the performance evaluation of large language models (LLMs), a new visualization technique called LLMMaps was proposed to assess the knowledge abilities of LLMs in different subfields in more detail. LLMMaps underwent evaluation through interviews and user feedback conducted in two separate studies involving medical researchers and language model developers. In the SciQ dataset, the performance of each model was assessed based on 997 questions, with ChatGPT achieving a performance of 92.68% (Puchert et al., 2023).

Studies have been conducted to develop various automated classifiers for cognitive levels of RBT in education using machine learning techniques. In one of the rule-based studies, the questions have been categorized with respect to the unrevised form of Bloom's taxonomy; preprocessing steps applied using techniques from natural language processing. This paper presents an approach that deals with overlapping verb keywords of Bloom's taxonomy, together with category weighting, and concludes that further rule development and testing are needed to enhance the system's effectiveness (Omar et al. 2012). In a study, using K-Nearest Neighbors, Logistic Regression, and Support Vector Machines, inquiries were classified by the cognitive domain of Bloom's taxonomy. The study used vectorization techniques such as TFPOS-IDF and word2vec for feature engineering. The W2V-TFPOSIDF and SVM techniques achieved an F1 score of 0.83 in one dataset and an F1 score of 0.89 in another dataset (Mohammed and Omar, 2020). Assessing the usability of the Naive Bayes classifier as a model for predicting exam questions according to the levels of Bloom's Cognitive Domain Taxonomy, researchers evaluated the efficacy of different vectorization methods using a dataset with 10-fold Cross-Validation. This dataset comprised 300 midterm and final exam questions, with feature extraction conducted using the TF-IDF method for classification. According to their evaluation, N-Gram TF-IDF achieved a Precision of 85%, and Words TF-IDF achieved a Recall of 82% (Aninditya et al., 2019). One of the works used pre-trained word embeddings, Long Short-Term Memory (LTSM), and a keyword/verb dictionary to represent the six levels of Bloom's taxonomy. The evaluation metrics revealed F1 scores of 74% for CLO classification and 87% for question classification (Shaikh et al., 2021). Utilizing Convolutional Neural Network (CNN) and LSTM for the classification of 844 questions about a Software Engineering course based on Bloom's Taxonomy, experiments revealed that the proposed CNN model achieved an accuracy rate of 80% in predicting the cognitive dimensions of Bloom's Taxonomy. Although the LSTM model was more successful during the training phase in predicting knowledge dimensions of Bloom's Taxonomy, during the testing phase, the CNN model demonstrated a higher accuracy rate (Laddha et al., 2021).

The objective of this study is to evaluate the question generation capability of three prominent chatbots, namely ChatGPT, Gemini, and Copilot, concerning their performance in classifying generations alignment with cognitive domain of Revised Bloom's Taxonomy.

Material and Methods

Artificial Intelligence Chatbots

Recently, AI chatbots have achieved accuracy in emulating complex language uses, opening numerous possibilities for applications across various fields, including education. AI chatbots such as ChatGPT, Gemini, and Microsoft Copilot exhibit significant potential in applications. However, the selection of appropriate AI chatbots must be based on specific application requirements and domains (Hiwa et al, 2024).

Chat-GPT, developed by OpenAI, is a language model. It employs a transformer architecture to understand and generate human-like responses based on the input it receives. (OpenAI, https://chat.openai.com/)

Gemini, initially introduced as Bard and built on the Pathways Language Model 2 (PaLM 2), is generate human-like responses to user queries. Integrated with Google's search network, it has access to up-to-date online information. (Google, https://gemini.google.com/app)

Microsoft Copilot, initially introduced as Bing Chat in a search engine and browser, has unified different chat services and has been integrated into an operating system with a dedicated key. It uses the model based on OpenAI's GPT-4 and operates on a freemium model. (Microsoft, https://copilot.microsoft.com/)

Datasets

The reference dataset includes Chris Drew's work (Drew, 2023), Ertekin's report (Ertekin, 2017), and the Ministry of National Education's environment courses curriculum gains (MEB, 2018), all within the scope of Bloom's Revised Taxonomy. The AI-generated dataset incorporates questions designed to align with the steps of RBT's cognitive domain. The question generation guidelines rely on definitions and verbs from the literature. Additionally, this dataset utilizes news topics from the Science Daily website (https://www.sciencedaily.com/). The site organizes news into 12 categories, and for this study, we select 14 topics from each category to ensure diversity.

Prompt Design

The definition and abilities of the categories/dimensions in preparing and classifying Turkish questions according to the Revised Bloom's Taxonomy is based on the work of Anderson and Krathwohl. According to the RBT, a common reference point in classification is verbs. Providing verbs to be used in Turkish questions in the prompt has ensured that the questions are more consistent. Verbs is referenced from the Turkish translations of the step descriptions in the literature and Stanny's compilation.

Prompt Design

- Yeniden Yapılandırılmış Bloom Taksonomisinin Bilişsel Süreç Basamağı [1 to 6]:
- [Name of the cognitive process category]
- [Definition of the cognitive process category]
- Bu basamağa uygun sorular üretmeni istiyorum.
- Öğretimde kullanılan dil gereği soruları emir kipiyle oluşturmalısın.
- Soruların bilişsel süreç basamağını ve uygun fiilleri içermesini sağlamalısın.
- Sorularda bu fiilleri kullanmalısın. "[category verbs]"
- Soru kelimelerini kullanmamalısın soruları verdiğim fillerle tamamlamalısın.
- Her konu için bir soru oluşturmalısın.
- Soruları tek fiil ile ifade etmelisin.
- Her soruda aynı fiili tekrarlamamaya çalış.
- Sorunun fiille anlam bütünlüğü oluşturmasına dikkat et.
- Soru ve konuları tablo formatında tut.
- Sütunlar: Soru, konu
- Soruların [..] basamağında olmasına dikkat et. [..] basamağı [düşük/yüksek] bilişsel seviyeli basamaktır. [category abilities] ve bu kapasiteleri sorularla ortaya çıkarmayı hedefler.
- Talimatlara uyarak her konudan birer soru oluşturacaksın.
- [Topics]

Figure 1. Prompt design for preparing and classifying Turkish questions according to the Revised Bloom's Taxonomy.

Vector Representation Method

The sentence vector is the numeric representation of words in a sentence. Obtaining this representation is done by text-embedding techniques. In this study, the pre-trained Turkish BERT model, without fine-tuning, is used to obtain the vector representation of the input questions.

BERT (Bidirectional Encoder Representations from Transformers) is a deep learning model based on transformers used in the field of natural language processing. (Devlin et al., 2019). The Turkish BERT (BERTurk) model is a trained and customized version to perform better on texts in the Turkish language. A hybrid model with BERTurk has demonstrated its highest performance in sentiment analysis within Turkish question-answer systems (Tohma et al., 2023) Another study illustrates how BERT improves results across various Turkish NLP tasks—such as sentiment analysis, cyberbullying identification, text classification, emotion recognition, and spam detection—by reducing the need for extensive pre-processing compared to traditional machine learning methods (Kutlu et al., 2017; Ozcift et al., 2021).

Similarity Metrics

The method used for selecting a similarity metric depends on the characteristics and requirements of the text similarity problem. Text similarity is measured by calculating the distance length between the vector representations of texts. Cosine similarity is one of the most frequently used measures in this respect (Wang and Dong, 2020).

The cosine similarity metric calculates the similarity between two vectors based on the angle between those vectors in vector space. Mathematically, the cosine similarity provides a value between 0 (lowest) and 1 (highest), indicating the similarity based on the angle. The cosine similarity between two vectors is calculated using the following formula:

$$\text{cosine\_similarity}(A, B) = (A \cdot B) / (\|A\| * \|B\|)$$

Machine Learning

Machine learning is a field that enables computers to learn without being explicitly programmed (Samuel, 1959). It utilizes data in the execution of algorithms that iteratively improve performance. Selecting the appropriate algorithm depends on the specific characteristics of the dataset.

SVM is a kernel-based machine learning model for classification and regression tasks. Various kernel types are used to transform datasets into high-dimensional spaces in different ways (Cervantes et al, 2020). Depending on the characteristics of the dataset, the kernel can be selected according to the class distribution. The kernel function used in this study:

Linear kernel: $K(xi, xj) = (xi * xj)$

Removing certain parts of the sentence

According to RBT, the parts of sentences containing verb and verbal are defining elements of the cognitive dimension for the classification task. The questions generated for our data are independent of the reference questions in terms of topic. The irrelevant (topic) parts of the sentences are filtered by determining parts of speech and dependency relations within the sentences according to specific linguistic criteria. For the method utilized the Spacy library adapted to the Turkish language (Altinok, 2023) such as Syntactic dependency of words -token.dep_, Part-of-speech of words - token.pos_.

Results

Distinguishing between AI-generated and expert-prepared questions can be challenging due to similarities in structure and potential ambiguity in language usage. To address this challenge, we carried out a comparison study that concentrated on the subtle differences in language and structural patterns between questions created by AI models and questions written by human experts.

Similarity Results

Utilizing a metric for comparing texts automates the measurement of similarity. In this context, the similarity metric method was used as an alternative analysis to compare the structural and semantic features of questions generated by AI models with those prepared by subject matter experts. The reference questions used in this section are drawn from the study by Mercan, which includes questions in each cognitive dimension on the same topic (Mercan, 2019).

Table 1. Cosine similarity metric comparison

References	Copilot	Gemini	ChatGPT
0	0.526	0.529	0.567
1	0.624	0.423	0.594
2	0.555	0.520	0.370
3	0.628	0.616	0.665
4	0.704	0.492	0.458
5	0.667	0.4785	0.328

Table 2. Samples of Turkish sentences reference sentences and generated sentences by AI tools according to Bloom's Taxonomy level

Bloom's Taxonomy Level	References	Copilot	Gemini	ChatGPT
Remember	Dünya’nın yuvarlak olduğu fikrini ilk ortaya atan kişi kim?	Dünyanın şeklinin temel bilgilerini hatırlayın.	Dünyanın şeklini ve katmanlarını tarif edin.	Dünyanın şeklini tanımlayın.
Understand	Dünyanın yuvarlaklığını kanıtlayan kişinin yöntemini açıklayın.	Dünyanın şeklini açıklayın ve yorumlayın.	Dünyanın katmanlarının özelliklerini ve ilişkilerini bir şemayla gösterin.	Dünyanın şeklini gösterin.
Apply	Dünyanın yuvarlak olmasıyla ilgilenen bilim insanlarının kronolojisini hazırlayın.	Dünyanın şeklini somut bir örnekle gösterin.	Dünyanın katmanlarını gösteren bir model maket oluşturun.	Dünya haritasında bir rota belirleyin ve neden seçtiğinizi açıklayın.
Analyze	Dünyanın şekli ve evrendeki yeri ile ilgili farklı düşünceleri karşılaştırarak benzerlikleri ve farklılıkları yorumlayın.	Dünyanın şeklini oluşturan bileşenleri ayırın ve nasıl bir araya geldiklerini inceleyin.	Dünyanın katmanlarının özelliklerini ve ilişkilerini bir diyagramda yeniden oluşturarak analiz edin.	Dünya haritasındaki farklı coğrafi bölgeleri karşılaştırın ve özelliklerini belirleyin.
Evaluate	Dünyanın şekli ve evrendeki yeri ile geçmişten günümüze gelen bilgi birikiminin yansımalarını değerlendirin.	Dünyanın şekliyle ilgili önemli özellikleri değerlendirin ve karşılaştırın.	Dünyanın katmanlarının özelliklerini ve ilişkilerini gösteren bir modelin eksikliklerini bulun ve öneriler sunun.	Dünya haritasındaki farklı coğrafi bölgelerin avantaj ve dezavantajlarını değerlendirin.
Create	Dünyanın yuvarlak olduğunu kanıtlamanız gerekirse günümüzde hangi yolları kullanırdınız?	Dünyanın şekli hakkında özgün bir açıklama veya model oluşturun.	Dünyanın katmanlarını gösteren ve interaktif bir model oluşturun.	Dünya haritasındaki farklı coğrafi bölgelerde sürdürülebilir kalkınma projeleri planlayın.

According to the results of Table 1, Gemini (0.423 to 0.616) and ChatGPT (0.328 to 0.665) showed low to moderate similarity to the questions prepared by experts and sometimes clearly deviated from the expert questions, suggesting significant variability and differences from the questions prepared by experts. As for the Copilot results, its showed moderate to high similarity (0.526 to 0.704), suggesting greater congruence in structure and meaning. These findings provide an empirical example of AI-generated questions can be distinguished from those prepared by experts based on these metrics, highlighting that even for the cognitive process of 'analysis' which achieves the highest average score, these AI models are unreliable sources to reference. Samples of Turkish sentences reference sentences and generated sentences by AI tools according to Bloom's Taxonomy level are shown Table 2.

Classification Results

In continuation of the study, classification analysis was performed using machine learning algorithms. 168 examples are generated for each RBT cognitive domain class. BERT vectors of the sentences served as input, with the cognitive domain serving as labels. The training phase employed the SVM algorithm with a linear kernel and a "one-vs-one" (ovo) decision function shape. In the classification using reference questions as test data, one hundred percent of the generated data was used for training. In the self-classification of the generated data, eighty percent of the data was used for training. A filtering approach with SpaCy was applied to classify the reference questions. This approach is applied to avoid topic-oriented classification. In contrast, the questions generated by artificial intelligence were classified using entire sentences without such filtering, each generation evaluated within itself.

Key steps in the classification process included: Tokenizing sentences and converting them into BERT embeddings. Classifying the resulting vectors using the SVM algorithm. Evaluating the performance of the algorithms using metrics such as accuracy, F1 score, precision, recall (as shown Table 3).
Definitions of the metrics are as follows:
Precision: TP / (TP + FP)
Recall (Sensitivity): TP / (TP + FN)
F1 Score: 2 * (precision * recall) / (precision + recall)
TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

Table 3. Classification of questions generated by AI tools

Bloom's Taxonomy Level	Copilot			Gemini			ChatGPT		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score	Precision	Recall	F1 Score
Remember	0.914	0.941	0.928	0.655	0.559	0.603	0.824	0.824	0.824
Understand	0.788	0.788	0.788	0.586	0.515	0.548	0.750	0.818	0.783
Apply	0.786	0.647	0.710	0.794	0.794	0.794	0.806	0.735	0.769
Analyze	0.806	0.879	0.841	0.667	0.788	0.722	0.784	0.879	0.829
Evaluate	0.914	0.941	0.928	0.722	0.765	0.743	0.969	0.912	0.939
Create	0.914	0.941	0.928	0.914	0.941	0.928	0.938	0.882	0.909

Based on the experimental results, Copilot achieved an accuracy of 0.856. The classification report indicates that the "remember," "evaluation," and "creation" classes all attained the highest F1 score of 0.928, whereas the "application" class had the lowest F1 score of 0.710. Similarly, ChatGPT achieved an accuracy of 0.841. Its classification report reveals that the "evaluation" class achieved the highest F1 score of 0.939, whereas the "application" class recorded the lowest F1 score of 0.769. In contrast, Gemini achieved an accuracy of 0.727. According to its classification report, the "creation" class obtained the highest F1 score of 0.928, while the "understanding" class showed the lowest F1 score of 0.548.

On average, the creation class shows the highest performance (0.921), while the understanding class exhibits the lowest (0.706) (as shown Table 4). This difference can be attributed to the fact that there is some overlap in the types of content such as fillers and functional elements between the ‘understanding’ class and the other classes. When cognitive levels are approached outside the framework determined by Bloom's taxonomy and not scientifically, 'understanding' often emerges as a populist concept that covers all cognitive processes. Consequently, these dynamics indirectly influence LLMs, which are shaped by human production and dominated by human knowledge.

Table 4. Classification of reference questions with generated questions

Bloom's Taxonomy Level	Copilot			Gemini			ChatGPT		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score	Precision	Recall	F1 Score
Remember	0.914	0.941	0.928	0.655	0.559	0.603	0.824	0.824	0.824
Understand	0.788	0.788	0.788	0.586	0.515	0.548	0.750	0.818	0.783
Apply	0.786	0.647	0.710	0.794	0.794	0.794	0.806	0.735	0.769
Analyze	0.806	0.879	0.841	0.667	0.788	0.722	0.784	0.879	0.829
Evaluate	0.914	0.941	0.928	0.722	0.765	0.743	0.969	0.912	0.939
Create	0.914	0.941	0.928	0.914	0.941	0.928	0.938	0.882	0.909

Based on the experimental results from evaluating the reference questions, the model trained with data generated by Copilot achieved an accuracy of 0.518. The classification report indicates that the 'creation' class attained the highest F1 score of 0.745, while the 'evaluation' class had the lowest F1 score of 0.303. In contrast, the model trained with data generated by ChatGPT achieved an accuracy of 0.512. According to its classification report, the "evaluation" class achieved the highest F1 score of 0.698, whereas the "application" class recorded the lowest F1 score of 0.255. Similarly, the model trained with data generated by Gemini achieved an accuracy of 0.512. Its classification report reveals that the "recall" class obtained the highest F1 score of 0.702, while the "analysis" class showed the lowest F1 score of 0.346.

On average, the "creation" class shows the highest performance (0.745), while the "evaluation" class exhibits the lowest (0.303). This difference may stem from the limited scope of verbs associated with the 'evaluation' class, necessitating attention to nuanced aspects beyond the verb within the remainder of the sentence. As a result, AI generations tend to focus predominantly on the 'verb' and the extracted 'topic,' potentially overlooking broader contextual nuances.

The analyses show the nuanced performance variations among AI-chatbot generations in executing cognitive tasks aligned with educational frameworks. Copilot demonstrates the highest overall performance among the AI-chatbots evaluated.

Discussion

In this paper, we have experimented with categorizing AI-generated questions in the educational domain, focusing on alignment with Revised Bloom's Taxonomy (RBT). Our analysis used AI chatbots; ChatGPT, Gemini, and Microsoft Copilot, each known for its text generation functions in the context of extensive applications. Our methodological approach involved creating an AI-generated dataset aligned with Bloom's Taxonomy cognitive processes using outputs from these chatbots. We compare this dataset with a reference dataset containing expert questions derived from scientific curriculum sources and educational literature. For structural similarity and semantic comparison of AI-generated versus expert questions, we used the Bert to obtain sentence vectors. The similarity between certain vectors is calculated using cosine similarity as the metric. Applying SVM as a machine learning classifier allowed further insights into the scaling analysis of all vectors.

In the results, while chatbots like Copilot are relatively promising in terms of their potential to generate educational questions according to cognitive frameworks, consideration would still have to be given to selection amongst AI models according to the specific educational targets. In this way, educators and curriculum developers have at their disposal a way to provide facilitated assistance to learners; however, they need to understand that these systems generate content that cannot capture the expected goals and nuances.

The study faces several limitations, including the size of the dataset, the specificity of the Turkish language, and the rapid evolution of AI models. Future research might add to these guidelines by exploring increased training methodologies of the AI models' alignment with educational frameworks and acceptability toward AI-generated educational materials. Additionally, incorporating diverse datasets and evaluating models in various linguistic and temporal settings may enhance the applicability of the findings.

In conclusion, this research explores the potential and limitations of using AI-generated questions in education and compares human-artificial intelligence language production. The results of the experimental study indicated that caution is necessary when applying AI technologies in educational contexts. Although AIs can extract specific patterns from data when generating human-like texts, they often exhibit a uniform pattern and lack nuanced context. As Chomsky’s study notes in his research, these patterns may remain superficial and dubious compared to human intelligence because they are not limited by rationality.

Conflict of Interest

The authors declare that they have no competing interests.

Ethical Approval Statements

Local Ethics Committee Approval was not obtained because experimental animals were not used in this study.

Data Availability Statement

The data used in the present study are available upon request from the corresponding author.

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